CATERPILLAR®

DEALER BEST PRACTICE SERIES

Component Life Tracking a Valuable CRC Performance Indicator

Application | Maintenance | Site Management | **Component Rebuild** | Safety | MARC Management

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1.0 Introduction

Virtually all dealer Component Rebuild Centers have a set of key performance metrics that are monitored as performance measures and indicators. This data is used to make management decisions needed to achieve business goals.

Dealers who have active MARC contracts supporting Mining customers understand the importance of achieving target component life. Meeting component life objectives is vital to the financial health of a MARC. Most CRC management teams would consider the key measure of rebuild quality from the CRC to be “Service Re-Do” or “Service Warranty”. While this an important indicator to monitor and manage, it is only a part of the component life issue. In the eyes of the Mining customer, cost per ton is a top critical success requirement. Mechanical cost per ton is driven by component life more than by any other factor. A process to track component life after rebuild is a valuable tool for managing CRC rebuild processes and MARC component life objectives.

2.0 Best Practice Description

It can be argued that “Service Re-Do” is a measure of CRC rebuild reliability, while “component life” is a measure of CRC rebuild durability. In a mining business environment where customer buying decisions are based on cost per hour (and often involve MARC contracts), delivering target component life is a critical success factor to the business. Tracking component life becomes a high priority in this environment.

Which customer fleets should be tracked?

- Component life cycles of all machines covered under MARC contracts.
- Component life cycles for any customer who sends all component rebuilds to the CRC.

What information should be tracked?

- Life cycle information for all drive train major components.
- Component serial number or tracking code.
- Life cycle - number of times component has been rebuilt
- Hours since last rebuild (current life) and fuel burn information for engines
- Total hours on component (all rebuilds)
- Work order number
- Reason for rebuild (PCR / Failure)
- For failures – Root cause of failure

Management Reports (sample reports below)

- Average component life cycle - by component / minesite
- Before / After failure ratio - by component/ minesite
- Component History
- Special analysis – histogram and Weibull

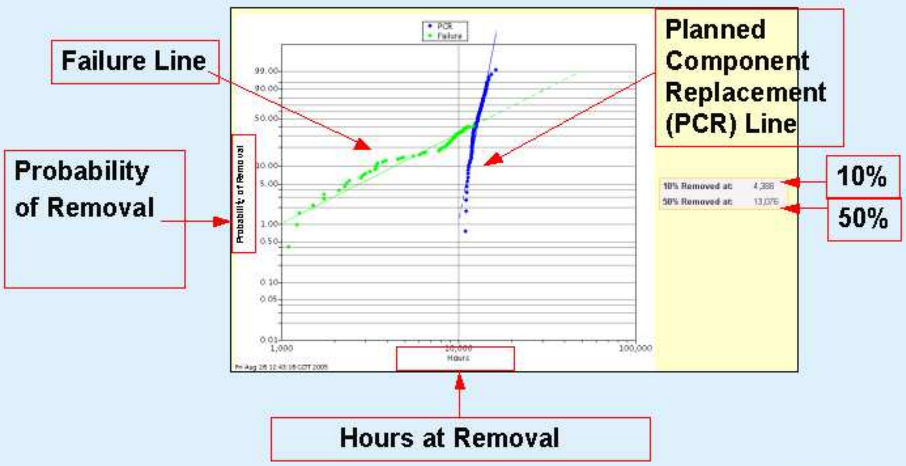
Life Cycle Summary

Dealer / Group:	NACD	**Report Date:**	08-Jun-2006
Component Type:	Wheel Group	**Sales Model Number:**	793C
Job Site Type:	All	**Job Site:**	All
Life Cycles:	All	**Installation Type:**	All
Removal Type:	All	**Removal Date:**	All

Printable View

Life	Avg. Component Life		Removal Reason							Actual Sample Size
			Failure		Wearout		PCR		Unknown	
	Percent	Avg. Hours	Percent	Avg. Hours	Percent	Avg. Hours	Percent	Avg. Hours	Percent	
1	100	14,261	33%	5,298	0%	0	62%	18,622	5%	143
2	100	11,408	40%	6,084	0%	0	53%	14,739	7%	62
3	100	10,545	40%	4,264	0%	0	54%	14,397	6%	42
4	100	8,798	61%	4,945	0%	0	36%	14,568	3%	36
5	100	7,704	68%	4,791	0%	0	31%	14,113	1%	16
6	100	6,567	80%	5,438	0%	0	20%	11,083	0%	5
7	100	7,752	71%	5,621	0%	0	28%	13,080	1%	7
8	100	9,472	50%	5,916	0%	0	50%	13,029	0%	2
9	100	7,413	50%	1,029	0%	0	50%	13,798	0%	4
10	100	1,867	50%	-7,529	0%	0	50%	11,263	0%	2

Appendix: MCTS Weibull Plot



* Three minimum samples required to complete the MCTS Weibull Plot

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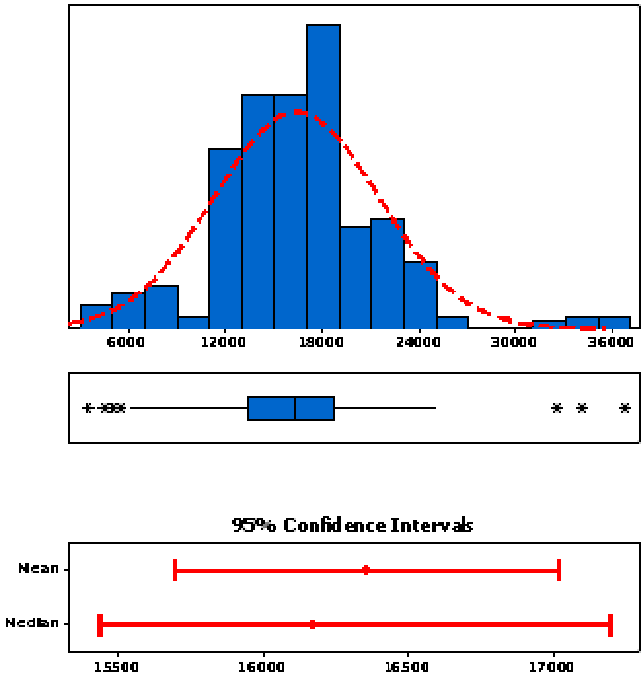
793C Worldwide Wheel Group PCR Life



Anderson-Darling Normality Test	
A-Squared	2.06
P-Value <	0.005
Mean	16354
StDev	5102
Variance	26034652
Skewness	0.66375
Kurtosis	2.70836
N	231
Minimum	3463
1st Quartile	13395
Median	16169
3rd Quartile	18696
Maximum	36698
95% Confidence Interval for Mean	
15693	17016
95% Confidence Interval for Median	
15436	17197
95% Confidence Interval for StDev	
4676	5616

Benchmarks

The commonly accepted benchmark for rebuild life is that it should be greater than 80% of the original new component life. The world-class performance benchmark is over 90% of new component life.

Machine major component life is affected by many variables, including design life, rebuild processes, application load and operator practices, and maintenance factors. The combined impact of all these variables determine component life. By tracking component life, especially for machines under maintenance contracts, we are able to better identify potential concerns and manage component life.

3.0 Implementation Steps

Utilize the Caterpillar Major Component Tracking System (MCTS)

- Provides a consistent worldwide method for dealers to track major components from cradle-to-grave through all life cycles.
- Provides one safe source for component life data for dealers and Caterpillar.
- Acts as a durability database by providing a complete list of all component lives from new to present.
- Provides actual and complete component life data, which is essential to accurately forecast component life and rebuild frequency when MARC contract costs are being calculated.

- Assists in daily fleet management decisions by proving instant, complete and accurate component history.
- Permits participating dealers to share component life data in the same format.
- Provides numerous, powerful, and user-friendly data analysis tools and reports.
- Use is free of charge to participating dealers. Caterpillar pays for system maintenance and the cost of tools and enhancements.

Dealers interested in using MCTS should contact:

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MCTS.cat.com

4.0 Benefits

- MARC contract management is improved by managing component life.
- MARC contract development has lower risk when actual component life data is available.
- CRC parts reuse and other rebuild processes are driven by a component life target benchmark
- CRC rebuild life can be a marketing tool to sell service, MARCs, and new machines.
- Component life data forms the basis for CRC continuous improvement initiatives.

5.0 Resources Required

- Time required to load historical component life data into the system.
- Manpower to enter data and keep current.

6.0 Supporting Attachments/ References

None.

7.0 Related Best Practices

N/A

8.0 Acknowledgements

This Best Practice was written by:
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